

ACADEMIC YEAR

2023-25

**COURSE STRUCTURE
& SYLLABUS**



**M.Sc. (Ag.) Agronomy
SCHOOL OF AGRICULTURE**

**GANDHI INSTITUTE OF ENGINEERING AND
TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR**

www.giet.edu

Agriculture

Program Educational Objectives:

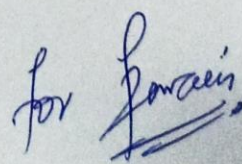
1. To produce manpower in the field of agriculture and allied sector.
2. To strengthen sustainable agriculture in the tribal area through development of innovative technologies.
3. To boost up Agri-entrepreneurial mindset amongst pass outs.
4. To develop problem solving mindset within the students for local farmers.
5. To provide solution to locality based problems.

Program Outcomes

1. Program produces quality graduates in the field of agriculture every year to cater to the needs of the state in agriculture and allied sector.
2. Faculties and Students are engaged in different research activities aiming at developing of eco-friendly & Cost-effective technologies supporting sustainable agriculture in the region.
3. Pass out are motivated and encouraged to establish their own agri-enterprise.
4. Students on acquiring village level experience make suitable interventions for farmer specific problems.
5. Research-Extension linkage is strengthened for facilitating dissemination of technology to farmer's field.
6. Hands on practical training is imparted which enables students through learning by doing.

Programme specific Outcome:

1. Addressing poor employability of the graduates emphasis on conservation
2. Develop climate resilient technologies
3. Innovative research through external funding
4. Imparting skill based training to farm stakeholders.
5. Making students aware about agriculture linked village activities
6. Provide diagnostic and remedial knowledge on real field situation
7. Acquaint students with ongoing agriculture related Govt. policies.
8. Enable student to acquire enterprise management ability.
9. Exposing students to industrial environment.
10. Imparting analytical skills to work independently
11. To develop leadership quality amongst students


Dean
School of Agriculture
GIET University, Gunupur

12. Encourage students to exhibit their creative, innovative ideas.
13. Special emphasis on Hands on training to gaining practical experience.
14. Involving students in team-sharing work experience
15. Developing & demonstrating 1 ha. ideal Integrated Farming System Unit.
16. Acquainting students on steps for establishing their own agriculture enterprise.
17. Use of Smart technologies in Agriculture.

**GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA,
GUNUPUR**

M.Sc. (Agronomy), SCHOOL OF AGRICULTURE

S.No	Semester	Subject Code	Subject Name	Credits
1	1st	AGRON 501	Modern Concepts in Crop Production	3(3+0)
2	1st	AGRON 502	Principles and Practices of Soil Fertility and Nutrient Management	3(2+1)
3	1st	AGRON 503	Principles and Practices of Weed Management	3(2+1)
4	1st	AGRON 504	Principles and Practices of Water Management	3(2+1)
5	1st	SOIL 511	Management of Problematic Soils and Water	2(1+1)
6	1st	PGS 502	Skills	1(0+1)
7	1st	PGS 503	Intellectual Property and Its Management in Agriculture	1(1+0)
			Total	16(11+5)

S.No	Semester	Subject Code	Subject Name	Credits
1	2nd	AGRON 507	Agronomy of oilseed, fibre and sugar crops	3(2+1)
2	2nd	AGRON 512	Dryland Farming and Watershed Management	3(2+1)
3	2nd	CPH 501	Stress Physiology	3(2+1)
4	2nd	STAT 511	Experimental Designs	3(2+1)
5	2nd	PGS 504	Basic Concepts in Laboratory Techniques	1(0+1)
			Total	13(8+5)

S.No	Semester	Subject Code	Subject Name	Credits
1	3rd	AGRON 506	Agronomy of Major Cereal and Pulses	2(2+0)
2	3rd	SOIL 508	Soil, water and air pollution	3(2+1)
3	3rd	MCA 511	Technologies, Computer Networking and Internet	2(1+1)
4	3rd	PGS 506	Agricultural Research, Research ethics and Rural Development Programmes	1(1+0)
5	3rd	AGRON 550	Master's Seminar	1(1+0)
6	3rd	AGRON 560	Master's Research	10(0+10)
			Total	19(7+12)

S.No	Semester	Subject Code	Subject Name	Credits
1	4th	MCA-512	Information Technology in Agriculture	2(2+0)
2	4th	PGS- 501	Library and Information Services	1(0+1)
3	4th	AGRON 560	Master's Research	20(0+20)
			Total	23(2+21)

M.Sc. in Agronomy

Major courses:

Course Title: Modern Concepts in Crop Production

Course Code: Agron 501

Credit Hours: 3+0

Course Outcomes

In this course, students will master contemporary agricultural techniques such as precision farming, biotechnology, and advanced irrigation systems. They will learn to evaluate and improve crop management practices for better productivity and sustainability, assess the environmental impacts of different methods, and implement integrated pest management strategies. Additionally, students will utilize data analysis to make informed decisions, enhancing overall crop production efficiency.

Program Outcomes

Students will possess advanced technical knowledge of cutting-edge agricultural technologies and methodologies. They will demonstrate strong problem-solving skills, advocate for sustainable practices, and engage in research and innovation to advance the field. Effective communication and collaboration with stakeholders, along with a commitment to professional ethics and responsibility, will be key aspects of their expertise in crop production.

Aim of the course: To teach the basic concepts of soil management and crop production.

Theory

Unit I

Crop growth analysis in relation to environment; geo-ecological zones of India.

Unit II

Quantitative agro-biological principles and inverse yield nitrogen law; Mitscherlich yield equation, its interpretation and applicability; Baule unit.

Unit III

Effect of lodging in cereals; physiology of grain yield in cereals; optimization of plant population and planting geometry in relation to different resources, concept of ideal plant type and crop modeling for desired crop yield.

Unit IV

Scientific principles of crop production; crop response production functions; concept of soil plant relations; yield and environmental stress, use of growth hormones and regulators for better adaptation in stressed condition.

Unit V

Integrated farming systems, organic farming, and resource conservation technology including modern concept of tillage; dry farming; determining the nutrient needs for yield potentiality of crop plants, concept of balance nutrition and integrated nutrient management; precision agriculture. Modern crop production concepts: soil less cultivation, Aeroponic, Hydroponic, Robotic and terrace farming. use of GIS, GPS and remote sensing in modern agriculture, precision farming and protected agriculture.

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment and class discussion

Learning outcome

Basic knowledge on soil management and crop production

Suggested Readings

- Balasubramaniyan P and Palaniappan SP. 2001. *Principles and Practices of Agronomy*. Agrobios.
- Fageria NK. 1992. *Maximizing Crop Yields*. Marcel Dekker.
- Havlin JL, Beaton JD, Tisdale SL and Nelson WL. 2006. *Soil Fertility and Fertilizers*. 7th Restructured and Revised Syllabi of Post-graduate Programmes Vol. 2 Ed. Prentice Hall.
- Paroda R.S. 2003. *Sustaining our Food Security*. Konark Publ.
- Reddy SR. 2000. *Principles of Crop Production*. Kalyani Publ.
- Sankaran S and Mudaliar TVS. 1997. *Principles of Agronomy*. The Bangalore Printing & Publ.
- Singh SS. 2006. *Principles and Practices of Agronomy*. Kalyani.
- Alvin PT and kozlowski TT (ed.). 1976. *Ecophysiology of Tropical Crops*. Academia Pul., New York.
- Gardner PP, Pearce GR and Mitchell RL. 1985. *Physiology of Crop Plants*. Scientific Pub. Jodhpur.
- Lal R. 1989. *Conservation tillage for sustainable agriculture: Tropics versus Temperate Environments*. *Advances in Agronomy* 42: 85-197.
- Wilsie CP. 1961. *Crop Adaptation and Distribution*. Euresia Pub., New Delhi.

Course Title : Principal and Practices of Soil Fertility and Nutrient Management**Course Code : Agron 502****Credit Hours : 2+1****Course Outcomes**

Ability to understand soil properties, assess soil fertility through testing, develop and implement effective nutrient management plans, apply appropriate fertilization techniques, and address issues related to soil degradation.

Program Outcomes

Involve applying core scientific principles to solve complex soil and nutrient management problems, designing and analyzing experiments, promoting sustainable agricultural practices, effectively communicating technical information, and demonstrating ethical and professional behavior while working collaboratively in multidisciplinary teams.

Aim of the course

To impart knowledge of fertilizers and manures as sources of plant nutrients and apprise about the integrated approach of plant nutrition and sustainability of soil fertility.

Theory**Unit I**

Soil fertility and productivity - factors affecting; features of good soil management; problems of supply and availability of nutrients; relation between nutrient supply and crop growth; organic farming - basic concepts and definitions.

Unit II

Criteria of essentiality of nutrients; Essential plant nutrients – their functions, nutrient deficiency symptoms; transformation and dynamics of major plant nutrients.

Unit III

Preparation and use of farmyard manure, compost, green manures, vermicompost, biofertilizers and other organic concentrates their composition, availability and crop responses; recycling of organic wastes and residue management. Soil less cultivation.

Unit IV

Commercial fertilizers; composition, relative fertilizer value and cost; crop response to different nutrients, residual effects and fertilizer use efficiency; agronomic, chemical and physiological, fertilizer mixtures and grades; methods of increasing fertilizer use efficiency; nutrient interactions.

Unit V

Time and methods of manures and fertilizers application; foliar application and its concept; relative performance of organic and inorganic nutrients; economics of fertilizer use; integrated nutrient management; use of vermincompost and residue wastes in crops.

Practical

- Determination of soil pH and soil EC
- Determination of soil organic C
- Determination of available N, P, K and S of soil
- Determination of total N, P, K and S of soil
- Determination of total N, P, K, S in plant
- Computation of optimum and economic yield

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment and class discussion

Learning outcome

Basic knowledge on soil fertility and management

Suggested Reading

- Brady NC and Weil RR. 2002. *The Nature and Properties of Soils*. 13th Ed. Pearson Edu.
- Fageria NK, Baligar VC and Jones CA. 1991. *Growth and Mineral Nutrition of Field Crops*. Marcel Dekker.
- Havlin JL, Beaton JD, Tisdale SL and Nelson WL. 2006. *Soil Fertility and Fertilizers*. 7th Ed. Prentice Hall.
- Prasad R and Power JF. 1997. *Soil Fertility Management for Sustainable Agriculture*. CRC Press.
- Yawalkar KS, Agrawal JP and Bokde S. 2000. *Manures and Fertilizers*. Agri-Horti Publ.

Course Title : Principles and Practices of Weed Management

Course Code : Agron 503

Credit Hours : 2+1

Course Outcomes

The ability to identify common weed species, understand their biology and ecology, assess the impact of weeds on crop production, and develop and implement integrated weed management strategies using cultural, mechanical, biological, and chemical methods.

Program Outcomes

Involve applying scientific knowledge to address complex weed management issues, conducting and interpreting research and experiments, developing sustainable and effective weed control practices, communicating findings and strategies clearly, and demonstrating ethical and professional behavior in managing agricultural resources.

Aim of the course

To familiarize the students about the weeds, herbicides and methods of weed control.

Theory

Weed biology, and ecology and classification, crop-weed competition including allelopathy; principles and methods of weed control and classification management; weed indices, weed shift in different eco-systems

Unit II

Herbicides introduction and history of their development; classification based on chemical, physiological application and selectivity; mode and mechanism of action of herbicides.

Unit III

Herbicide structure - activity relationship; factors affecting the efficiency of herbicides; herbicide formulations, herbicide mixtures, sequential application of herbicides, rotation; weed control through use of nano-herbicides and bio-herbicides, myco-herbicides bio-agents, and allelochemicals; movement of herbicides in soil and plant, Degradation of herbicides in soil and plants; herbicide resistance, residue, persistence and management; development of herbicide resistance in weeds and crops and their management, herbicide combination and rotation.

Unit IV

Weed management in major crops and cropping systems; alien, invasive and parasitic weeds and their management; weed shifts in cropping systems; aquatic and perennial weed control; weed control in non-crop area.

Unit V

Integrated weed management; recent development in weed management- robotics, use of drones and aeroplanes, organic etc., cost: benefit analysis of weed management.

Practical

- Identification of important weeds of different crops, Preparation of a weed herbarium, Weed survey in crops and cropping systems, Crop-weed competition studies, Weed indices calculation and interpretation with data, Preparation of spray solutions of herbicides for high and low-volume

sprayers, Use of various types of spray pumps and nozzles and calculation of swath width, Economics of weed control, Herbicide resistance analysis in plant and soil,

- Bioassay of herbicide resistance residues,
- Calculation of herbicidal herbicide requirement

Teaching methods/activities

Classroom teaching with AV aids, group discussion, field visit to identify weeds.

Learning outcome

Basic knowledge on weed identification and control for crop production

Suggested Reading

- Böger, Peter, Wakabayashi, Ko, Hirai, Kenji (Eds.). 2002. *Herbicide Classes in Development. Mode of Action, Targets, Genetic Engineering, Chemistry*. Springer.
- Chauhan B and Mahajan G. 2014. *Recent Advances in Weed Management*. Springer.
- Das TK. 2008. *Weed Science: Basics and Applications*, Jain Brothers (New Delhi).
- Fennimore, Steven A and Bell, Carl. 2014. *Principles of Weed Control*, 4th Ed, California Weed Sci. Soc.
- Gupta OP. 2007. *Weed Management: Principles and Practices*, 2nd Ed.
- Jugulan, Mithila (ed). 2017. *Biology, Physiology and Molecular Biology of Weeds*. CRC Press
- Monaco TJ, Weller SC and Ashton FM. 2014. *Weed Science Principles and Practices*, Wiley
- Powles SB and Shaner DL. 2001. *Herbicide Resistance and World Grains*, CRC Press.
- Walia US. 2006. *Weed Management*, Kalyani.
- Zimdahl RL. (ed). 2018. *Integrated Weed Management for Sustainable Agriculture*, B. D. Sci. Pub.

Course Title : Principles and Practices of Water Management

Course Code : Agron 504

Credit Hours : 2+1

Course Outcomes

The ability to understand and analyze water cycles and availability, assess irrigation needs and systems, design and implement efficient water management practices, and evaluate the impact of water management on agricultural productivity and environmental sustainability.

Program Outcomes

Involve applying fundamental scientific knowledge to solve complex water management problems, conducting and interpreting experiments related to water use and irrigation, developing sustainable water management strategies, effectively communicating technical information, and demonstrating ethical and professional practices in managing water resources.

Aim of the course

To teach the principles of water management and practices to enhance the water productivity]

Theory

Unit I

Water and its role in plants; Irrigation: Definition and objectives, water resources and irrigation development in of India and concerned state, major irrigation projects, extent of area and crops irrigated in India and in different states.

Unit II

Field water cycle, water movement in soil and plants; transpiration; soil-waterplant relationships; water absorption by plants; plant response to water stress, crop plant adaptation to moisture stress condition. Water availability and its relationship with nutrient availability and losses.

Unit III

Soil, plant and meteorological factors determining water needs of crops, scheduling, depth and methods of irrigation; micro irrigation systems; deficit irrigation; fertigation; management of water in controlled environments and polyhouses. Irrigation efficiency and water use efficiency.

Unit IV

Water management of crop and cropping system, Quality of irrigation water and management of saline water for irrigation, water use efficiency, Crop water requirement- estimation of ET and effective rainfall; Water management of the major crops and cropping systems. Automated irrigation system.

Unit V

Excess of soil water and plant growth; water management in problem soils, drainage requirement of crops and methods of field drainage, their layout and spacing; rain water management and its utilization for crop production.

Unit VI

Quality of irrigation water and management of saline water for irrigation, water management in problem soils

Unit VII

Soil moisture conservation, water harvesting, rain water management and its utilization for crop production.

Unit VIII

Hydroponics,

Unit IX

Water management of crops under climate change scenario.

Practical

- Determination of Field capacity by field method
- Determination of Permanent Wilting Point by sunflower pot culture technique
- Determination of Field capacity and Permanent Wilting Point by Pressure Plate Apparatus
- Determination of Hygroscopic Coefficient
- Determination of maximum water holding capacity of soil
- Measurement of matric potential using gauge and mercury type tensiometer
- Determination of soil-moisture characteristics curves
- Determination of saturated hydraulic conductivity by constant and falling head method

- Determination of hydraulic conductivity of saturated soil below the water table by auger hole method
- Measurement of soil water diffusivity
- Estimation of unsaturated hydraulic conductivity
- Estimation of upward flux of water using tensiometer and from depth ground water table
- Determination of irrigation requirement of crops (calculations)
- Determination of effective rainfall (calculations)
- Determination of ET of crops by soil moisture depletion method¹⁶. Determination of water requirements of crops
- Measurement of irrigation water by volume and velocity-area method
- Measurement of irrigation water by measuring devices and calculation of irrigation efficiency
- Determination of infiltration rate by double ring infiltrometer

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment and field visit

Learning outcome

Basic knowledge on water management for optimization of crop yield

Suggested Reading

- Majumdar DK. 2014. *Irrigation Water Management: Principles and Practice*. PHL Learning private publishers
- Mukund Joshi. 2013. *A Text Book of Irrigation and Water Management Hardcover*, Kalyani publishers
- Lenka D. 1999. *Irrigation and Drainage*. Kalyani.
- Michael AM. 1978. *Irrigation: Theory and Practice*. Vikas Publ.
- Paliwal KV. 1972. *Irrigation with Saline Water*. IARI Monograph, New Delhi.
- Panda SC. 2003. *Principles and Practices of Water Management*. Agrobios.
- Prihar SS and Sandhu BS. 1987. *Irrigation of Food Crops - Principles and Practices*. ICAR.
- Reddy SR. 2000. *Principles of Crop Production*. Kalyani.
- Singh Pratap and Maliwal PL. 2005. *Technologies for Food Security and Sustainable Agriculture*. Agrotech Publ.

Course Title : Agronomy of Major Cereals and Pulses

Course Code : Agron 506

Credit Hours : 2+0

Course Outcomes

The ability to identify and understand the growth requirements of major cereal and pulse crops, apply agronomic practices for optimal production, manage crop rotation and soil fertility specific to these crops, and evaluate the impact of various cultivation techniques on yield and quality.

Program Outcomes

Involve applying scientific principles to solve agronomic problems related to cereal and pulse crops, conducting and analyzing experiments on crop performance, developing and implementing

sustainable agronomic practices, communicating technical information effectively, and demonstrating ethical and professional behavior in agricultural practices.

Aim of the course

To impart knowledge of crop husbandry of cereals and pulse crops.

Theory

Origin and history, area and production, classification, improved varieties, adaptability, climate, soil, water and cultural requirements, nutrition, quality components, handling and processing of the produce for maximum production of:

Unit I: *Rabi* cereals.

Unit II: *Kharif* cereals.

Unit III: *Rabi* pulses.

Unit IV: *Kharif* pulses.

Practical

- Phenological studies at different growth stages of crop
- Estimation of crop yield on the basis of yield attributes
- Formulation of cropping schemes for various farm sizes and calculation of cropping and rotational intensities
- Working out growth indices (CGR, RGR, NAR, LAI, LAD, LAR, LWR, SLA, SLW etc)
- Assessment of land use and yield advantage (Rotational intensity, Cropping intensity, Diversity Index, Sustainable Yield Index Crop Equivalent Yield, Land Equivalent ration, Aggressiveness, Relative Crowding Coefficient, Competition Ratio and ATER etc)
- Estimation of protein content in pulses
- Planning and layout of field experiments
- Judging of physiological maturity in different crops
- Intercultural operations in different crops
- Determination of cost of cultivation of different crops
- Working out harvest index of various crops
- Study of seed production techniques in selected crops
- Visit of field experiments on cultural, fertilizer, weed control and water management aspects
- Visit to nearby villages for identification of constraints in crop production

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment and class discussion

Learning outcome

Basic knowledge on cereals and pulse growing in the country .

Resources

- Das NR. 2007. *Introduction to Crops of India*. Scientific Publ.
- Hunsigi G and Krishna KR. 1998. *Science of Field Crop Production*. Oxford & IBH.
- Jeswani LM and Baldev B. 1997. *Advances in Pulse Production Technology*. ICAR.
- Khare D and Bhale MS. 2000. *Seed Technology*. Scientific Publ.
- Kumar Ranjeet and Singh NP. 2003. *Maize Production in India: Golden Grain in Transition*.

IARI, New Delhi.

- Pal M, Deka J and Rai RK. 1996. *Fundamentals of Cereal Crop Production*. Tata McGraw Hill.
- Prasad Rajendra. 2002. *Text Book of Field Crop Production*. ICAR.
- Singh C, Singh P and Singh R. 2003. *Modern Techniques of Raising Field Crops*. Oxford & IBH.
- Singh SS. 1998. *Crop Management*. Kalyani.
- Yadav DS. 1992. *Pulse Crops*. Kalyani.

Course Title : Agronomy of Oilseed, Fibre and Sugar Crops

Course Code : Agron 507

Credit Hours : 2+1

Course Outcomes

The ability to identify and understand the specific agronomic requirements of oilseed, fiber, and sugar crops; implement effective cultivation practices for each crop type; manage crop-specific challenges such as pest control and nutrient management; and evaluate the impact of different agronomic practices on crop yield and quality.

Program Outcomes

Involve applying scientific principles to address complex issues related to oilseed, fiber, and sugar crops, designing and interpreting experiments to improve crop management, developing and implementing sustainable practices for these crops, effectively communicating technical and research findings, and adhering to ethical and professional standards in agricultural practices.

Aim of the course

To teach the crop husbandry of oilseed, fiber and sugar crops

Theory

Origin and history, area and production, classification, improved varieties, adaptability, climate, soil, water and cultural requirements, nutrition, quality component, handling and processing of the produce for maximum production of:

Unit I

Rabi oilseeds – Rapeseed and mustard, Linseed and Niger

Unit II

Kharif oilseeds - Groundnut, Sesame, Castor, Sunflower, Soybean and Safflower

Unit III

Fiber crops - Cotton, Jute, Ramie and Mesta.

Unit IV

Sugar crops – Sugar-beet and Sugarcane.

Practical

- Planning and layout of field experiments
- Cutting of sugarcane setts, its treatment and methods of sowing, tying and propping of sugarcane
- Determination of cane maturity and calculation on purity percentage, recovery percentage and sucrose content in cane juice phenological studies at different growth stages of crop
- Intercultural operations in different crops

- Cotton seed treatment
- Working out growth indices (CGR, RGR, NAR, LAI, LAD, LAR, LWR, SLA, SLW etc)
- Assessment of land use and yield advantage (Rotational intensity, Cropping intensity, Diversity Index, Sustainable Yield Index Crop Equivalent Yield, Land Equivalent ration, Aggressiveness, Relative Crowding Coefficient, Competition Ratio and ATER etc)
- Judging of physiological maturity in different crops and working out harvest index
- Working out cost of cultivation of different crops
- Estimation of crop yield on the basis of yield attributes
- Formulation of cropping schemes for various farm sizes and calculation of cropping and rotational intensities
- Determination of oil content in oilseeds and computation of oil yield
- Estimation of quality of fibre of different fibre crops
- Study of seed production techniques in various crops
- Visit of field experiments on cultural, fertilizer, weed control and water management aspects
- Visit to nearby villages for identification of constraints in crop production

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment and class discussion

Learning outcome

Basic knowledge on production of oil seed, sugar and fibre crops.

Suggested Reading

- Das NR. 2007. *Introduction to Crops of India*. Scientific Publ.
- Das PC. 1997. *Oilseed Crops of India*. Kalyani.
- Lakshmikantam N. 1983. *Technology in Sugarcane Growing*. 2nd Ed. Oxford & IBH.
- Prasad Rajendra. 2002. *Text Book of Field Crop Production*. ICAR.
- Singh C, Singh P & Singh R. 2003. *Modern Techniques of Raising FieldCrops*. Oxford & IBH.
- Singh SS. 1998. *Crop Management*. Kalyani.

Course Title : Dryland Farming and Watershed Management

Course Code. : Agron 512

Credit Hours : 2+1

Course Outcomes

The ability to understand the principles of dryland farming, assess and manage water resources in arid and semi-arid regions, implement effective soil and water conservation techniques, and develop strategies for sustainable land use and watershed management.

Program Outcomes

Involve applying scientific principles to address challenges in dryland farming and watershed management, conducting and interpreting research to improve water and soil management practices, developing and promoting sustainable practices for arid and semi-arid regions, communicating technical findings effectively, and demonstrating ethical and professional behavior in managing natural resources.

Aim of the course

To teach the basic concepts and practices of dry land farming and soil moisture conservation.

Theory**Unit I**

Definition, concept and characteristics of dry land farming; dry land versus rainfed farming; significance and dimensions of dry land farming in Indian agriculture.

Unit II

Soil and climatic parameters with special emphasis on rainfall characteristics; constraints limiting crop production in dry land areas; types of drought, characterization of environment for water availability; crop planning for erratic and aberrant weather conditions.

Unit III

Stress physiology and resistance to drought, adaptation of crop plants to drought, drought management strategies; preparation of appropriate crop plans for dry land areas; mid contingent plan for aberrant weather conditions.

Unit IV

Tillage, tilth, frequency and depth of cultivation, compaction in soil tillage; concept of conservation tillage; tillage in relation to weed control and moisture conservation; techniques and practices of soil moisture conservation (use of mulches, kinds, effectiveness and economics); antitranspirants; soil and crop management techniques, seeding and efficient fertilizer use.

Unit V

Concept of watershed resource management, problems, approach and components.

Practical

- Method of Seed Priming
- Determination of moisture content of germination of important dryland crops
- Determination of Relative Water Content and Saturation Deficit of Leaf
- Moisture stress effects and recovery behaviour of important crops
- Estimation of Potential ET by Thornthwaite method
- Estimation of Reference ET by Penman Monteith Method
- Classification of climate by Thornthwaite method (based on moisture index, humidity index and aridity index)
- Classification of climate by Koppen Method
- Estimation of water balance by Thornthwaite method
- Estimation of water balance by FAO method
- Assessment of drought
- Estimation of length of growing period
- Estimation of probability of rain and crop planning for different drought condition
- Spray of anti-transpirants and their effect on crops
- Water use efficiency
- Visit to dryland research stations and watershed projects

Teaching methods/activities

Classroom teaching with AV aids, group discussion, assignment.

Learning outcome

Basic knowledge on dry land farming and soil moisture conservation.

IX. Suggested Reading

- Reddy TY. 2018. *Dryland Agriculture Principles and Practices*, Kalyani publishers
- Das NR. 2007. *Tillage and Crop Production*. Scientific Publ.
- Dhopte AM. 2002. *Agrotechnology for Dryland Farming*. Scientific Publ.
- Dhruv Narayan VV. 2002. *Soil and Water Conservation Research in India*. ICAR.
- Gupta US. (Ed.). 1995. *Production and Improvements of Crops for Drylands*. Oxford & IBH.
- Katyal JC and Farrington J. 1995. *Research for Rainfed Farming*. CRIDA.
- Rao SC and Ryan J. 2007. *Challenges and Strategies of Dryland Agriculture*. Scientific Publ.
- Singh P and Maliwal PL. 2005. *Technologies for Food Security and Sustainable Agriculture*. Agrotech Publ. Company.
- Singh RP. 1988. *Improved Agronomic Practices for Dryland Crops*. CRIDA.
- Singh RP. 2005. *Sustainable Development of Dryland Agriculture in India*. Scientific Publ.
- Singh SD. 1998. *Arid Land Irrigation and Ecological Management*. Scientific Publ.
- Venkateshwarlu J. 2004. *Rainfed Agriculture in India. Research and Development Scenario*. ICAR.

Minor courses

Course Title: Soil, Water and Air Pollution

Course Code: Soil 508

Credit Hours: 2+1

Course Outcomes

The ability to identify and assess sources and impacts of pollution on soil, water, and air; understand the mechanisms and effects of different pollutants; develop and evaluate strategies for pollution prevention and control; and apply remediation techniques to mitigate environmental contamination.

Program Outcomes

Involve applying scientific principles to address pollution issues, conducting and analyzing research to understand and manage pollution, developing effective strategies for pollution control and remediation, communicating scientific and technical information clearly, and adhering to ethical and professional standards in environmental management.

Aim of the course

To make the student aware of the problems of soil, water and air pollution associated with use of soils for crop production.

Theory

Unit I

Soil, water and air pollution problems associated with agriculture, nature and extent.

Unit II

Nature and sources of pollutants – agricultural, industrial, urban wastes, fertilizers and pesticides, acid rains, oil spills etc.; air, water and soil pollutants- their CPC standards and effect on plants, animals and human beings.

Unit III

Sewage and industrial effluents–their composition and effect on soil properties/ health, and plant growth and human beings; soil as sink for waste disposal.

Unit IV

Pesticides–their classification, behaviour in soil and effect on soil microorganisms.

Unit V

Toxic elements–their sources, behaviour in soils, effect on nutrients availability, effect on plant and human health.

Unit VI

Pollution of water resources due to leaching of nutrients and pesticides from soil; emission of greenhouse gases–carbon dioxide, methane and nitrous oxide.

Unit VII

Risk assessment of polluted soil, Remediation/ amelioration of contaminated soil and water; remote sensing applications in monitoring and management of soil and water pollution.

Practical

Sampling of sewage waters, sewage sludge, solid/ liquid industrial wastes, polluted soils and plants and their processing, Estimation of dissolved and suspended solids, chemical oxygen demand (COD), biological demand (BOD), measurement of coliform (MPN), nitrate and ammoniacal nitrogen and phosphorus, heavy metal content in effluents, Heavy metals in contaminated soils and plants, Management of contaminants in soil and plants to safe guard food safety, Air sampling and determination of particulate matter and oxides of sulphur, NO₂ and O₂ conc. Visit to various industrial sites to study the impact of pollutants on soil and plants.

Suggested Reading

- Lal R, Kimble J, Levine E and Stewart BA. 1995. Soil Management and Greenhouse Effect. CRC Press.
- Middlebrooks EJ. 1979. Industrial Pollution Control. Vol. I. Agro-Industries. John Wiley Interscience.
- Ross SM. Toxic Metals in Soil Plant Systems. John Wiley & Sons.
- Vesilund PA and Pierce 1983. Environmental Pollution and Control. Ann Arbor Science Publ.

Course Title: Management of Problem Soils and Water

Course Code: Soil 511

Credit Hours: 1+1

Course Outcomes

The ability to identify and characterize problem soils and water issues, understand their impact on agricultural productivity and environmental health, develop and implement management strategies to improve soil and water quality, and apply remediation techniques to address specific soil and water challenges.

Program Outcomes

Involve applying scientific principles to solve complex issues related to problem soils and water, conducting and interpreting research to develop effective management practices, creating sustainable solutions for soil and water improvement, communicating findings and strategies effectively, and demonstrating ethical and professional behavior in environmental and agricultural managem

Aim of the course

To educate students about basic concepts of problem soils and brackish water, and their management. Attention will be on management of problem soils and safe use of brackish water in relation to crop production.

Theory

Unit I

Area and distribution of problem soils—acidic, saline, sodic and physically degraded soils; origin and basic concept of problematic soils, and factors responsible.

Unit II

Morphological features of saline, sodic and saline-sodic soils; characterization of salt- affected soils-soluble salts, ESP, pH; physical, chemical and microbiological properties.

Unit III

Management of salt-affected soils; salt tolerance of crops- mechanism and ratings; salt stress meaning and its effect on crop growth, monitoring of soil salinity in the field; management principles for sandy, clayey, red lateritic and dryland soils.

Unit IV

Acid soils-nature of soil acidity, sources of soil acidity; effect on plant growth, lime requirement of acid soils; management of acid soils; biological sickness of soils and its management.

Unit V

Quality of irrigation water; management of brackish water for irrigation; salt balance under irrigation; characterization of brackish waters, area and extent; relationship in water use and quality.

Unit VI

Agronomic practices in relation to problematic soils; cropping pattern for utilizing poor quality ground waters.

Practical

Characterization of acid, acid sulfate, salt-affected and calcareous soils, Determination of cations (Na^+ , K^+ , Ca^{++} and Mg^{++}) in groundwater and soil samples, Determination of an ions (Cl^- , SO_4^- , CO_3^- and HCO_3^-) in ground waters and soil samples, Lime and gypsum requirements of acid and sodic soils.

Teaching methods/activities

Classroom teaching with AV aids, group discussion, oral presentation by students.

Learning outcome

Experience on solving field problem of problem soil and waters.

Resources

- Bear FE. 1964. Chemistry of the Soil. Oxford & IBH.
- Jurinak JJ. 1978. Salt-affected Soils. Department of Soil Science & Biometeorology. Utah State University
- USDA Handbook No. 60. 1954. Diagnosis and improvement of Saline and Alkali Soils. Oxford & IBH.

Course Title: Stress physiology

Course No. : CPH 501

Credit hours: 2+1

Course Outcomes

The ability to understand the physiological responses of organisms to various stressors, analyze the impact of stress on biological systems, evaluate mechanisms of stress adaptation and resilience, and apply knowledge to manage and mitigate stress effects in both natural and controlled environments.

Program Outcomes

Involve applying scientific principles to understand and address physiological responses to stress, conducting and interpreting research on stress adaptation, developing strategies to manage stress in

biological systems, communicating scientific findings effectively, and demonstrating ethical and professional conduct in stress physiology studies and applications.

Objectives: To study various types of stresses in crop production and strategies to overcome them.

Theory

UNIT I

Response of plants to abiotic stress: Abiotic stresses affecting plant productivity, interactions between biotic and abiotic stresses.

UNIT II

Drought characteristic features, water potential in the soil-plant-air continuum. Development of water deficits, energy balance concept.

UNIT III

Transpiration and its regulation stomatal functions.

UNIT IV

Physiological processes affected by drought. Drought resistance mechanism: Escape, dehydration postponement (drought avoidance), dehydration tolerance and characteristics of resurrection plants, osmotic adjustment, osmoprotectants, stress proteins, water use efficiency as a drought resistant trait. Water logging and flooding injury. Molecular responses to water deficit: expression of regulatory and functional genes and its significance.

UNIT V

Stress and determined hormones ABA as a signaling molecule Cytokinin as a negative signal. Oxidative stress: Relative Oxygen Species (ROS), role of scavenging systems (SOD catalase etc.).

UNIT VI

Stress due to high temperature: heat wave tolerance mechanisms- role of membrane lipids in high temperature tolerance, functions of HSP's.

UNIT VII

Low temperature stress: Chilling and freezing stress, frost and cold injury effects on physiological process, crucial role of membrane lipids.

UNIT VIII

Salinity: Species variation in salt tolerance, salinity effects at Cellular and whole plant level, tolerance mechanisms, salt tolerance in Glycophytes and halophytes.

UNIT IX

Heavy metal stress: Aluminum and cadmium toxicity in acid soils, role of phytochelatins (heavy metal binding proteins). Practical Determination of proline content of plant parts; determination of Relative Leaf water content of plants; Quantification of anti oxidative enzymes like Super oxide desmutase (SOD); Determination of membrane injury index (MII); Determination of chlorophyll stability index (CSI) ; Studying the effect of ABA / Cytokinin on stomatal behavior.

Learning Outcome: The students will understand various aspects of stress physiology such as physiological and molecular basis of abiotic stress tolerance in plants; the knowledge in stress physiology will be useful for developing climate resilient genotypes for sustainable crop production

Supporting courses:

Course Title: Experimental Designs

Course No.: STAT – 511

Credit Hour: 3(2+1)

Course Outcomes

The ability to understand and apply various experimental design principles such as randomization, replication, and blocking; construct and implement appropriate experimental designs for different research scenarios; analyze and interpret statistical data from experiments using appropriate techniques; and assess the effectiveness and validity of experimental designs.

Program Outcomes

Involve applying statistical principles to design and conduct experiments, utilizing advanced statistical methods to analyze and interpret data, developing robust research methodologies, effectively communicating statistical findings and insights, and adhering to ethical standards in statistical research and experimentation.

Objective: This course is meant for students of agricultural and animal sciences other than Agricultural Statistics. Designing an experiment is an integrated component of research in almost all sciences. The students would be exposed to concepts of Design of Experiments so as to enable them to understand the concepts involved in planning, designing their experiments and analysis of experimental data.

Theory

UNIT I

Need for designing of experiments, characteristics of a good design. Basic principles of designs randomization, replication and local control.

UNIT II

Uniformity trials, size and shape of plots and blocks, Analysis of variance, Completely randomized design, randomized block design and Latin square design.

UNIT III

Factorial experiments, (symmetrical as well as asymmetrical). Orthogonality and partitioning of degrees of freedom. Concept of confounding.

UNIT IV

Split plot and strip plot designs, analysis of covariance and missing plot techniques in randomized block and Latin square designs; Transformations, Balanced Incomplete Block Design, resolvable designs and their applications, Lattice design, alpha design - concepts, randomization procedure, analysis and interpretation of results. Response surfaces. Combined analysis.

Practical

Uniformity trial data analysis, formation of plots and blocks, Fairfield Smith Law, Analysis of data obtained from CRD, RBD, LSD, Analysis of factorial experiments; Analysis with missing data; Split plot and strip plot designs.

Course Title : Introduction to Networking and Internet Applications

Course Code : MCA 511

Credit Hours : 1+1

Course Outcomes

The ability to understand fundamental networking concepts and protocols, configure and manage basic network setups, utilize and apply various internet applications effectively, and troubleshoot common network issues.

Program Outcomes

Involve applying networking principles to design and implement network solutions, utilizing internet technologies to develop and manage applications, analyzing and solving network-related problems, effectively communicating technical information, and adhering to professional and ethical standards in networking and internet application development.

Aim of the course

This is a course on Introduction to Networking and Internet Applications that aims at exposing the students to understand Computer networking and web applications development.

Theory

Unit I

Networking fundamentals, types of networking, network topology; Introduction to File Transfer Protocol (FTP), Telnet, Simple Mail Transfer Protocol (SMTP), Internet Protocol v4 & v6. Network infrastructure and Security-switches, routers, firewall, intranet, internet, Virtual Private Network

Unit II

World Wide Web (www), working with Internet; Web pages, web sites, web servers; Web Applications.

Unit III

Hyper Text Markup Language (HTML), DHTML, web based application development. Static websites, dynamic websites. Client Side processing – scripting languages, JQuery. Server Side processing ASP.NET/JSP

VI. Practical

- Network and mail configuration;
- Using Network Services;
- Browsing of Internet;
- Creation of web pages;
- Creation of websites using HTML and scripting languages.

Suggested Reading

- Cox V, Wermers L and Reding E.E. 2006. *HTML Illustrated Complete*. 3rd Ed. Course Technology.
- Niederst J. 2001. *Web Design in a Nutshell*. O'Reilly Media.
- Tanenbaum A.S. 2003. *Computer Networks*. Prentice Hall of India.

Course Title: Information Technology in Agriculture**Course Code: MCA 512****Credit hours: 2(1+1)****Course Outcomes**

The ability to understand and apply IT tools and technologies to enhance agricultural practices, integrate information systems for managing farm data and operations, utilize precision agriculture techniques for improving crop yields and resource efficiency, and evaluate the impact of IT solutions on agricultural productivity and sustainability.

Program Outcomes

Involve applying IT principles to solve agricultural challenges, developing and implementing information systems and technologies for efficient farm management, analyzing and interpreting data to inform agricultural decisions, communicating technical solutions effectively, and demonstrating ethical and professional behavior in the application of IT to agriculture.

Unit-I

Introduction and Applications of e-Agriculture, Introduction to Online Agricultural resources: Consortium for e-resources in Agriculture (CeRA), e-agriculture community, Agriculture: National Portal of India. Agricultural Datasets and Databases: Agricola, Agris. Need of Biological databases in Agricultural Sciences.

Unit-II

Smartphone Apps in Agriculture for farm advisory, Weather forecasting, types, methods, tools & techniques, Use of ICT in Agriculture, Computer Models for understanding plant processes.

Unit-III

Crop models, concepts & techniques, types of crop models, spatial data and their management in GIS; Remote sensing concepts and application in agriculture, Global positioning system (GPS), components and its functions.

Unit-IV

Agricultural Journalism – Meaning, Scope and Importance, Sources of news, Kisan call centers, e-chaupal, RRA, PRA tools and techniques KVK, Adopter categories, MANAGE, EEI: extension education institute.

Practical: Uniformity trial data analysis, formation of plots and blocks, Analysis of data obtained from Completely Randomized Design, Randomized Block Design, Latin Square Design; Analysis of factorial experiments without and with confounding; Analysis with missing data; Split plot designs; Transformation of data; Fitting of response surfaces.

Suggested Readings:

- Agri Informatics: An Introduction (Industry Series), by R Chakravarthy, ICFAI University Press.
- E-Agriculture: Concepts and Applications (Agriculture Series), Rahul Gupta (Author), ICFAI University Press.
- Yadav, D S, Foundations of IT, New Age, Delhi.

- Introduction to Bioinformatics by Teresa Attwood, David Parry-Smith 1st edition; Prentice Hall publications
- Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins by Andreas D. Baxevanis and B. F. Francis Ouellette (Eds), 2nd Edition; Willey & Sons publications.
- Bioinformatics: Sequence, Structure, and Databanks: A Practical Approach by Des Higgins, Willie Taylor; OUP. BIOS Instant Notes in Bioinformatics by Charlie Hodgman, Andrew French, David Westhead, Taylor & Francis publishing; 2 edition

Common Courses:

I. Course Title: LIBRARY AND INFORMATION SERVICES

II. Course Code. : PGS- 501

III. Credit Hours: 0+1

Course Outcomes

The ability to understand and apply principles of library science and information management, effectively organize and manage library resources, assist users in locating and utilizing information, and evaluate and implement technologies to enhance library services and access.

Program Outcomes

Involve applying library and information science principles to manage and provide access to information resources, designing and implementing effective information retrieval systems, analyzing and responding to user needs, communicating information and technology solutions effectively, and adhering to ethical standards in information management and library services.

Objective

To equip the library users with skills to trace information from libraries efficiently, to apprise them of information and knowledge resources, to carry out literature survey, to formulate information search strategies, and to use modern tools (Internet, OPAC, search engines, etc.) of information search.

Practical

Introduction to library and its services; Role of libraries in education, research and technology transfer; Classification systems and organization of library; Sources of information- Primary Sources, Secondary Sources and Tertiary Sources; Intricacies of abstracting and indexing services (Science Citation Index, Biological Abstracts, Chemical Abstracts, CABI Abstracts, etc.); Tracing information from reference sources; Literature survey; Citation techniques/ Preparation of bibliography; Use of CD-ROM Databases, Online Public Access Catalogue and other computerized library services; Use of Internet including search engines and its resources; eresources access methods.

I. Course Title : TECHNICAL WRITING AND COMMUNICATIONS SKILLS

II. Course Code. : PGS-502

III. Credit Hours : 0+1

Course Outcomes

The ability to produce clear, concise, and well-organized technical documents, adapt communication strategies to different audiences and purposes, effectively use technical writing tools and formats, and critically review and revise written materials for accuracy and clarity.

Program Outcomes

Involve applying advanced writing and communication principles to create effective technical documentation, utilizing various communication channels and tools to convey information, analyzing and addressing the needs of diverse audiences, presenting information clearly and professionally, and demonstrating ethical and professional standards in technical communication.

Objective

To equip the students/ scholars with skills to write dissertations, research papers, etc. To equip the students/ scholars with skills to communicate and articulate in English (verbal as well as writing).

Practical (Technical Writing)

- Various forms of scientific writings- theses, technical papers, reviews, manuals, etc.;
- Various parts of thesis and research communications (title page, authorship contents page, preface, introduction, review of literature, material and methods, experimental results and discussion);
- Writing of abstracts, summaries, précis, citations, etc.;
- Commonly used abbreviations in the theses and research communications;
- Illustrations, photographs and drawings with suitable captions; pagination, numbering of tables and illustrations;
- Writing of numbers and dates in scientific write-ups;
- Editing and proof-reading;
- Writing of a review article;
- Communication Skills - Grammar (Tenses, parts of speech, clauses, punctuation marks);
- Error analysis (Common errors), Concord, Collocation, Phonetic symbols and transcription;
- Accentual pattern: Weak forms in connected speech;
- Participation in group discussion;
- Facing an interview;
- Presentation of scientific papers.

Suggested Readings

1. Barnes and Noble. Robert C. (Ed.). 2005. *Spoken English: Flourish Your Language*.
2. *Chicago Manual of Style*. 14th Ed. 1996. Prentice Hall of India.
3. *Collins' Cobuild English Dictionary*. 1995.
4. Harper Collins. Gordon HM and Walter JA. 1970. *Technical Writing*. 3rd Ed.
5. Holt, Rinehart and Winston. Hornby AS. 2000. *Comp. Oxford Advanced Learner's Dictionary of Current English*. 6th Ed. Oxford University Press.
6. James HS. 1994. *Handbook for Technical Writing*. NTC Business Books.
7. Joseph G. 2000. *MLA Handbook for Writers of Research Papers*. 5th Ed. Affiliated East-West Press.
8. Mohan K. 2005. *Speaking English Effectively*. MacMillan India.
9. Richard WS. 1969. *Technical Writing*.
10. Sethi J and Dhamija PV. 2004. *Course in Phonetics and Spoken English*. 2nd Ed. Prentice Hall of India.
11. Wren PC and Martin H. 2006. *High School English Grammar and Composition*. S. Chand & Co.

I. Course Title: INTELLECTUAL PROPERTY AND ITS MANAGEMENT IN AGRICULTURE

II. Course Code: PGS 503

III. Credit Hours: 1+0

Course Outcomes

The ability to understand various forms of intellectual property (IP) relevant to agriculture, such as patents, trademarks, and copyrights; apply IP laws and regulations to agricultural innovations; develop strategies for protecting and managing IP in agricultural research and practice; and evaluate the impact of IP management on agricultural development and commercialization.

Program Outcomes

Involve applying knowledge of IP laws and management principles to protect agricultural innovations, developing and implementing effective IP strategies, analyzing the implications of IP on agricultural research and industry, communicating IP-related information effectively, and adhering to ethical standards in the management and commercialization of agricultural intellectual property.

Objective

The main objective of this course is to equip students and stakeholders with knowledge of Intellectual Property Rights (IPR) related protection systems, their significance and use of IPR as a tool for wealth and value creation in a knowledge based economy.

Theory

Historical perspectives and need for the introduction of Intellectual Property Right regime; TRIPs and various provisions in TRIPS Agreement; Intellectual Property and Intellectual Property Rights (IPR), benefits of securing IPRs; Indian Legislations for the protection of various types of Intellectual Properties; Fundamentals of patents, copyrights, geographical indications, designs and layout, trade secrets and traditional knowledge, trademarks, protection of plant varieties and farmers' rights and biodiversity protection; Protectable subject matters, protection in biotechnology, protection of other biological materials, ownership and period of protection; National Biodiversity protection initiatives; Convention on Biological Diversity; International Treaty on Plant Genetic Resources for Food and Agriculture; Licensing of technologies, Material transfer agreements, Research collaboration Agreement, License Agreement.

Suggested Readings

1. Erbisch FH and Maredia K.1998. *Intellectual Property Rights in Agricultural Biotechnology*. CABI.
 2. Ganguli P. 2001. *Intellectual Property Rights: Unleashing Knowledge Economy*. McGraw-Hill.
 3. *Intellectual Property Rights: Key to New Wealth Generation*. 2001. NRDC and Aesthetic Technologies.
 4. Ministry of Agriculture, Government of India. 2004. *State of Indian Farmer*. Vol. V. Technology Generation and IPR Issues. Academic Foundation.
 5. Rothschild M and Scott N. (Ed.). 2003. *Intellectual Property Rights in Animal Breeding and Genetics*. CABI.
 6. Saha R. (Ed.). 2006. *Intellectual Property Rights in NAM and Other Developing Countries: A Compendium on Law and Policies*. Daya Publ. House.
- The Indian Acts - Patents Act, 1970 and amendments; Design Act, 2000; Trademarks

Act, 1999; The Copyright Act, 1957 and amendments; Layout Design Act, 2000; PPV and FR Act 2001, and Rules 2003; The Biological Diversity Act, 2002.

I. Course Title : BASIC CONCEPTS IN LABORATORY TECHNIQUES

II. Course Code. : PGS 504

III. Credit Hours : 0+1

Course Outcomes

The ability to understand fundamental laboratory techniques and equipment, accurately perform standard laboratory procedures, adhere to safety protocols and best practices, and analyze and interpret experimental data effectively.

Program Outcomes

Involve applying foundational laboratory principles to conduct experiments, utilizing laboratory techniques and equipment efficiently, ensuring adherence to safety and quality standards, interpreting and communicating scientific data, and demonstrating professional and ethical behavior in laboratory settings.

Objective

To acquaint the students about the basics of commonly used techniques in laboratory.

Practical

- Safety measures while in Lab;
- Handling of chemical substances;
- Use of burettes, pipettes, measuring cylinders, flasks, separatory funnel, condensers, micropipettes and vaccupets;
- Washing, drying and sterilization of glassware;
- Drying of solvents/ chemicals;
- Weighing and preparation of solutions of different strengths and their dilution;
- Handling techniques of solutions;
- Preparation of different agro-chemical doses in field and pot applications;
- Preparation of solutions of acids;
- Neutralisation of acid and bases;
- Preparation of buffers of different strengths and pH values;
- Use and handling of microscope, laminar flow, vacuum pumps, viscometer, thermometer, magnetic stirrer, micro-ovens, incubators, sandbath, waterbath, oilbath;
- Electric wiring and earthing;
- Preparation of media and methods of sterilization;
- Seed viability testing, testing of pollen viability;
- Tissue culture of crop plants;
- Description of flowering plants in botanical terms in relation to taxonomy.

Suggested Readings

1. Furr AK. 2000. *CRC Hand Book of Laboratory Safety*. CRC Press.
2. Gabb MH and Latchem WE. 1968. *A Handbook of Laboratory Solutions*. Chemical Publ. Co.

I. Course Title : AGRICULTURAL RESEARCH, RESEARCH ETHICS AND RURAL DEVELOPMENT PROGRAMMES

II. Course Code. : PGS 506

III. Credit Hours : 1+0

Course Outcomes

Include the ability to conduct rigorous agricultural research, apply ethical principles in research practices, design and implement effective rural development programs, and evaluate the impact of research and development initiatives on rural communities.

Program Outcomes

Involve applying research methodologies and ethical standards to address agricultural and rural development challenges, designing and evaluating programs that improve rural livelihoods, conducting and analyzing research to inform policy and practice, communicating research findings effectively, and demonstrating ethical and professional conduct in research and development activities.

Objective

To enlighten the students about the organization and functioning of agricultural research systems at national and international levels, research ethics, and rural development programmes and policies of Government.

Theory

UNIT I History of agriculture in brief; Global agricultural research system: need, scope, opportunities; Role in promoting food security, reducing poverty and protecting the environment; National Agricultural Research Systems (NARS) and Regional Agricultural Research Institutions; Consultative Group on International Agricultural Research (CGIAR): International Agricultural Research Centres (IARC), partnership with NARS, role as a partner in the global agricultural research system, strengthening capacities at national and regional levels; International fellowships for scientific mobility.

UNIT II Research ethics: research integrity, research safety in laboratories, welfare of animals used in research, computer ethics, standards and problems in research ethics.

UNIT III Concept and connotations of rural development, rural development policies and strategies. Rural development programmes: Community Development Programme, Intensive Agricultural District Programme, Special group – Area Specific Programme, Integrated Rural Development Programme (IRDP) Panchayati Raj Institutions, Co-operatives, Voluntary Agencies/ Non-Governmental Organisations. Critical evaluation of rural development policies and programmes. Constraints in implementation of rural policies and programmes.

Suggested Readings

1. Bhalla GS and Singh G. 2001. *Indian Agriculture - Four Decades of Development*. Sage Publ.
2. Punia MS. *Manual on International Research and Research Ethics*. CCS Haryana Agricultural University, Hisar.
3. Rao BSV. 2007. *Rural Development Strategies and Role of Institutions - Issues, Innovations and Initiatives*. Mittal Publ.
4. Singh K. 1998. *Rural Development - Principles, Policies and Management*. Sage Publ.

for Perain

Dean

School of Agriculture
GIET University, Gunupur